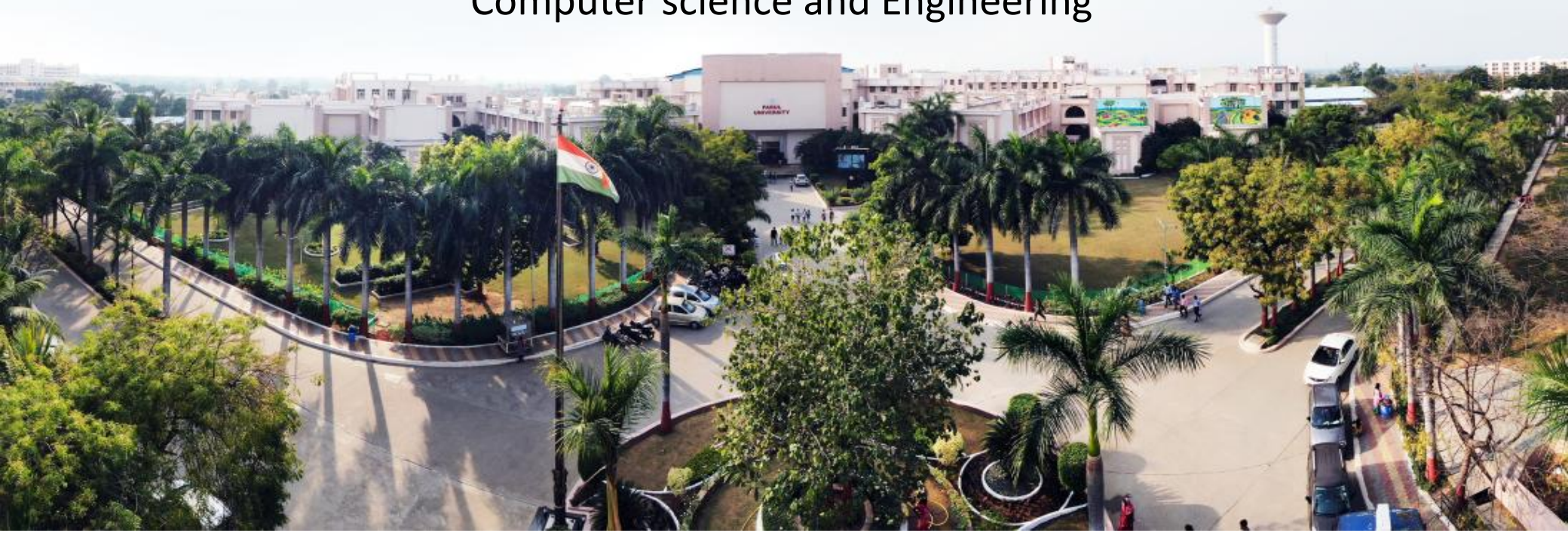




Image Processing (303105381)

Mr. Anurag Kewat

Lecturer
Computer science and Engineering



UNIT-8

Object Recognition

Outline

1. What is Object Recognition?
2. Patterns and Pattern Classes
3. Decision-Theoretic Methods
4. Structural Methods

What is Object Recognition?

The process of identifying and classifying objects in images or data by assigning them to predefined categories.

Core Task: Given an input pattern (image, signal), determine which class it belongs to from a set of known classes.

Applications: Face recognition, medical imaging, autonomous vehicles, industrial inspection, document analysis.

Two Paradigms: Statistical (Decision-Theoretic) — uses probability & features; Structural — uses shape grammar & relationships.

Patterns and Pattern Classes : Core Concepts

Pattern: A quantitative or structural description of an object — may be a vector of features $x = [x_1, x_2, \dots, x_n]$.

Feature: Measurable property: color histogram, edge gradient, texture, area, perimeter, HOG descriptor.

Pattern Class: A set of patterns sharing similar characteristics; represented as a cluster in feature space.

Feature Space: n-dimensional space where each axis is a feature; patterns appear as points or vectors.

Patterns and Pattern Classes : Types of Patterns

Numeric/Feature Vectors : Fixed-length vectors — classically used in statistical classifiers (e.g., pixel intensities, SIFT).

String / Sequence : Ordered sequence of primitives; used in OCR, speech, 1-D signals (e.g., abcba).

Graph / Relational : Nodes represent parts; edges represent relationships — used in structural recognition.

Hierarchical : Tree structures capturing part-whole decomposition of complex objects.

Feature Spaces & Distance Metrics

Patterns are represented as points in a multi-dimensional feature space. Similarity is measured by distance.

Euclidean Distance : Most common; works when features have equal scale.

$$d(x, y) = \sqrt{\sum (x_i - y_i)^2}$$

Mahalanobis Distance : Accounts for covariance; scale-invariant.

$$d = \sqrt{(x - \mu)^T \Sigma^{-1} (x - \mu)}$$

Feature Spaces & Distance Metrics

City-Block (L1) : Robust to outliers; used in image features.

$$d(x, y) = \sum |x_i - y_i|$$

Hamming Distance : For binary strings; used in code comparison.

$$d = \# \text{ positions differ}$$

Cosine Similarity : Direction-based; used in text & high-dim spaces.

$$\cos \theta = (x \cdot y) / (|x| |y|)$$

Edit Distance : String patterns; used in sequence recognition.

$$\min \text{ insert/delete/replace}$$

Decision-Theoretic Methods – Overview

Use statistical features and mathematical decision rules to assign patterns to classes.



Decision-theoretic methods are pattern recognition techniques that classify objects or patterns by using statistical decision rules and probability theory to minimize classification errors and make optimal decisions based on available data.

Decision-Theoretic Methods – Overview

Use statistical features and mathematical decision rules to assign patterns to classes.

Classifier	Principle	Strengths	Limitations
Bayesian	Posterior probability $P(C_i x)$	Optimal if distributions known	Needs density estimation
k-NN	Majority vote among k nearest	Non-parametric, intuitive	Slow for large datasets
Linear Discriminant	Hyperplane boundary $wx+b=0$	Simple, interpretable	Fails for non-linear data
SVM	Max-margin hyperplane	Works in high dims, robust	Kernel choice, slow train
Neural Network	Learned deep features	Handles complex patterns	Black-box, data hungry

Bayesian / Statistical Classifier

Bayes' Theorem:

$$P(C_i \mid x) = [P(x \mid C_i) \times P(C_i)] / P(x)$$

$P(C_i \mid x)$: Posterior probability — probability of class C_i given observed pattern x

$P(x \mid C_i)$: Likelihood — probability of observing x given class C_i (class-conditional density)

$P(C_i)$: Prior probability — frequency of class C_i in the training set

$P(x)$: Evidence — normalizing constant; same for all classes

Decision Rule: Assign x to class C_i that maximizes $P(C_i \mid x) \rightarrow$ Minimum Error Rate Classifier

Discriminant Functions & Decision Boundaries

Definition : $g_i(x)$ assigns a score to each class C_i . Pattern x is assigned to class with highest score.

Linear Form : $g_i(x) = w_i^T x + w_{i0}$

w = weight vector, w_0 = threshold/bias

Decision Rule : Assign to C_i if $g_i(x) > g_j(x)$ for all $j \neq i$

Boundary : $g_i(x) = g_j(x)$ defines the decision surface between classes C_i and C_j

Types of Decision Boundaries

Linear : Hyperplane — works when classes are linearly separable (Perceptron, LDA).

Quadratic : Conic sections (ellipse, parabola) — when classes have different covariance matrices.

Piecewise Linear : Multiple linear segments — used in k-NN and decision trees.

Non-linear / Kernel : Complex surfaces via kernel functions in SVM; $\phi(x)$ maps to higher dimension.

k-Nearest Neighbor

- Stores all training samples — no explicit model training.
- For a test point x , find the k nearest training points by distance.
- Assign x to the majority class among those k neighbors.
- $k=1$: Nearest Neighbor rule — simplest case.
- Error rate $\leq 2 \times$ Bayes error for large n (Cover & Hart, 1967).
- Drawback: $O(n)$ storage & search cost; slow at test time.

Support Vector Machine

- Finds the optimal hyperplane that maximizes the margin between classes.
- Support vectors: training points closest to the boundary — define the margin.
- Margin = $2 / ||w||$ — maximized by minimizing $||w||^2$.
- Soft-margin SVM (C-parameter) allows some misclassifications for noise.
- Kernel Trick: $K(x_i, x_j) = \phi(x_i) \cdot \phi(x_j)$ — maps to higher-dim space.
- Common kernels: Linear, Polynomial, Radial Basis Function (RBF/Gaussian).

Structural Methods – Overview

Represent patterns as symbolic structures (strings, trees, graphs) and use formal grammars or graph matching for recognition.

Primitives & Relations:

- Primitives: basic structural elements (line segments, curves, angles, blobs).
- Relations: spatial/topological connections between primitives (above, adjacent, connected).
- Pattern = arrangement of primitives + their relationships.

Structural Methods – Overview

Syntax-Based (String Grammar) :

- Patterns described as strings of primitive symbols.
- Formal grammar $G = (N, T, P, S)$: non-terminals, terminals, production rules, start symbol.
- A pattern is accepted if the string is in the grammar's language $L(G)$.

Structural Methods – Overview

Tree Grammars :

- Hierarchical patterns (nested structures) represented as trees.
- Tree grammar rules: parent → children compositions.
- Used for 2-D patterns, scene parsing, shape decomposition.

Graph-Based Methods :

- Patterns = attributed relational graphs (ARG): nodes with attributes, edges with relations.
- Recognition via graph isomorphism or subgraph matching.
- Used for 3-D object recognition, chemical structures, scene analysis.

String Grammar & Syntactic Pattern Recognition

Formal Grammar: $G = (N, T, P, S)$

$N = \{\text{non-terminal symbols (abstract categories)}\}$

$T = \{\text{terminal symbols (primitives/alphabet)}\}$

$P = \{\text{production rules: } \alpha \rightarrow \beta\}$

$S = \text{start symbol}$

Chomsky Hierarchy — Grammar Types

Parsing: Given a string (pattern), determine if it belongs to $L(G)$. Accepted = recognized as that class.

Type 0 — Unrestricted

No restriction on rules; most powerful

Type 1 — Context-Sensitive $\alpha A \beta \rightarrow \alpha \gamma \beta$; used for complex structures**Type 2 — Context-Free (CFG)** $A \rightarrow \gamma$; very common in pattern recognition**Type 3 — Regular** $A \rightarrow aB$ or $A \rightarrow a$; simplest; finite automata

Graph-Based Structural Recognition

Attributed Relational Graphs (ARG) :

- Nodes:** Represent object parts or regions with attribute vectors (size, color, shape).
- Edges:** Encode relationships between parts (adjacency, above, inside, angle).
- Matching:** Compare test graph with model graph using graph isomorphism algorithms.
- Subgraph Match:** Find if model graph is a subgraph of the scene graph.
- Error-Tolerant:** Use cost functions for inexact matching to handle noise & distortion.

Graph-Based Structural Recognition

Applications & Structural vs. Statistical :

- 3D object recognition from CAD models
- Fingerprint matching (minutiae graph)
- Molecular structure classification
- Scene understanding & spatial reasoning
- Handwriting & symbol recognition

Key Advantage: Captures shape topology, not just numerical features — handles variability and structural distortions naturally.

Statistical vs. Structural Methods – Comparison

Aspect	Statistical / Decision-Theoretic	Statistical / Decision-Theoretic
Pattern Rep.	Feature vectors (numeric)	Strings, trees, graphs (symbolic)
Knowledge	Statistical distributions / margins	Structural rules / grammar
Classifier	Bayes, k-NN, SVM, Neural Net	Parser, graph matcher
Strengths	Handles noisy, real-valued data	Captures shape & topology explicitly
Weaknesses	Loses structural information	Computationally expensive matching
Training	Statistical learning from examples	Grammar inference / manual design
Applications	Image classification, face ID, speech	Shape analysis, 3D, document parsing

Summary

- **Patterns & Pattern Classes** : Patterns are feature vectors or structures. Classes are clusters in feature space. Distance metrics define similarity.
- **Decision-Theoretic Methods** : Use statistical properties of data. Bayes classifier minimizes error by maximizing posterior. k-NN, LDA, SVM & neural networks offer different decision boundary complexities.
- **Discriminant Functions** : $g_i(x)$ scores each class. Decision boundaries are surfaces where scores are equal. Linear boundaries = hyperplanes; non-linear = kernel methods.
- **Structural Methods** : Represent patterns as strings, trees, or graphs. Formal grammars parse strings. Graph matching handles 2D/3D shape recognition using relational structures.



Parul[®] University

